

Quality of Life and Diabetes

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Summary

Quality of life is an important health outcome in its own right, representing the ultimate goal of all health interventions. This paper reviews the published, English-language literature on self-perceived quality of life among adults with diabetes. Quality of life is measured as physical and social functioning, and perceived physical and mental well-being. People with diabetes have a worse quality of life than people with no chronic illness, but a better quality of life than people with most other serious chronic diseases. Duration and type of diabetes are not consistently associated with quality of life. Intensive treatment does not impair quality of life, and having better glycemic control is associated with better quality of life. Complications of diabetes are the most important disease-specific determinant of quality of life. Numerous demographic and psychosocial factors influence quality of life and should be controlled when comparing subgroups. Studies of clinical and educational interventions suggest that improving patients' health status and perceived ability to control their disease results in improved quality of life. Methodologically, it is important to use multidimensional assessments of quality of life, and to include both generic and disease-specific measures. Quality of life measures should be used to guide and evaluate treatment interventions. Copyright © 1999 John Wiley & Sons, Ltd.

Keywords quality of life; diabetes; chronic disease; well-being

Introduction

Why are quality of life issues important in diabetes?

In 1948 the World Health Organization defined health from a new perspective, stating that health was defined not only by the absence of disease and infirmity, but also by the presence of physical, mental and social well-being [1]. In recent years, there has been a burgeoning interest in quality of life issues, and especially in health-related quality of life, fueled by several factors, including a growing body of evidence concerning the potent effect of psychosocial factors on physical health outcomes, and dramatic changes in the organization and delivery of health care.

People with diabetes often feel challenged by their disease and its day-to-day management demands. And these demands are substantial. Patients must deal with their diabetes all day, every day, making countless decisions in an often futile effort to approximate the non-diabetic metabolic state. Diabetes therapy, such as taking insulin, can substantially affect quality of life either positively, by reducing symptoms of high blood sugar, for instance, or negatively, by increasing symptoms of low blood sugar, for example. The psychosocial toll of living with diabetes is often a heavy one, and this toll can often, in turn, affect self-care behavior and, ultimately, long-term glycemic control, the risk of developing long-term complications, and quality of life.

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There is now good evidence that psychosocial issues are critical to good diabetes care [2,3]. Psychosocial factors often determine self-management behaviors, and psychosocial variables (such as depression) are often stronger predictors of medical outcomes such as hospitalization and mortality than are physiologic and metabolic measures (such as the presence of complications, BMI and HbA_{1c}) [4].

Recent developments in the fields of health outcome research and health technology assessment have also fueled the tremendous increase in the use of quality of life evaluation as a technique for clinical research [5,6]. Greater attention is now being devoted to evaluating the quality of health care and the economic value associated with new interventions. Managed care organizations have stimulated a growing effort to determine whether the costs associated with new or existing therapies and educational interventions are justified within fairly short time frames, often less than 3 years. As a consequence, researchers and health plan administrators are focusing on shorter-term patient outcomes, including functional health status, satisfaction with health care, and overall quality of life. The goals of monitoring psychosocial well-being and quality of life in people with diabetes include: identification of patients who are depressed or anxious; evaluating new treatments by identifying psychological costs and benefits; and identifying dissatisfaction with treatment and other aspects of care

Purpose of this paper

This paper addresses a number of issues related to quality of life in people with diabetes (see Figure 1). We describe the range of factors considered by various researchers to be elements of quality of life and identify the major instruments used to measure quality of life issues in people with diabetes. We then examine the factors associated with quality of life among persons with diabetes and compare their quality of life to persons who do not have diabetes. Finally, we examine interventions that may have effects on quality of life in diabetes. In the course of this review we consider a number of methodological issues which face the field.

The paper examines several categories of factors related to quality of life:

1. disease-specific *medical* predictors, including type and duration of diabetes, treatment regimen, level of glycemic control and the presence of complications
2. diabetes-specific *attitudinal* predictors, including diabetes self-efficacy, locus of control and social support
3. *demographic* predictors, including gender, education, ethnicity, age and marital status.

It should be noted that we use the term 'predictors' to describe the relationship between any of the factors mentioned and quality of life only when we conceive of quality of life as an outcome. It is clear that the true causal relationships among all of the identified variables are complex and often reciprocal. Thus, for example, quality

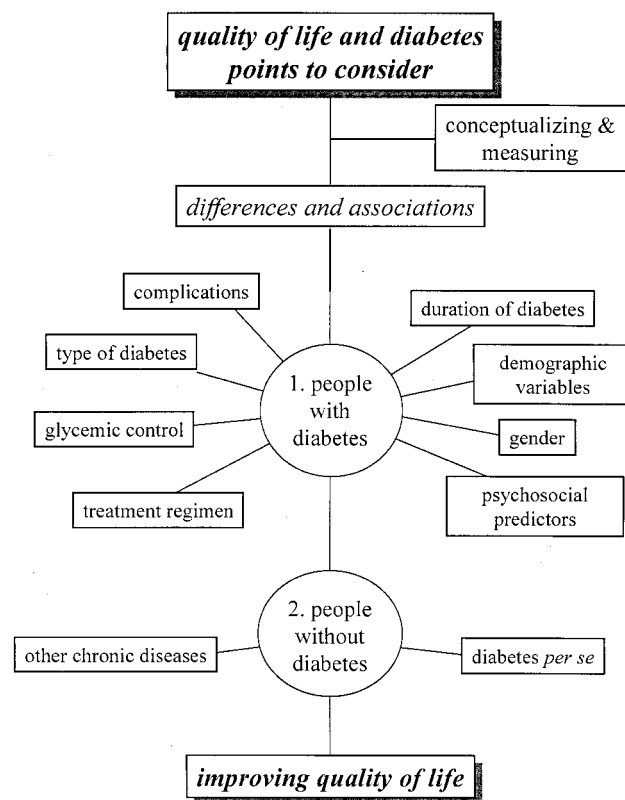


Figure 1. Outline

of life may affect diabetes self-efficacy, self-care behavior, glycemic control and complications, just as each of these latter variables may affect each other and quality of life.

The paper reviews published English-language empirical studies identified through a systematic literature search of Medline and PsychLit. The keywords used as search criteria were 'quality of life' and 'diabetes'. We excluded all studies which dealt with children or family members of diabetic patients, and studies in which persons with diabetes were combined with patients having other disorders (but without diabetes) into a single 'chronic disease' group. Snowball sampling was used to include studies cited in articles which were identified. We also included relevant articles known to us through other sources. We did not obtain an exhaustive inventory of relevant research, nor did we eliminate studies based on methodological criteria. The studies reviewed represent the full range of studies in the literature on quality of life and diabetes.

The studies reviewed range widely in terms of their purpose, design, population of interest, and the way in which they define quality of life. The designs best-represented in the literature are:

1. *cross-sectional or non-randomized longitudinal studies*, designed to identify predictors or correlates of quality of life
2. *randomized studies of clinical interventions*, including studies comparing medications and insulin delivery devices, and studies of educational interventions which incorporate assessments of quality of life.

How is quality of life conceptualized and measured?

In this paper we take quality of life to be a multi-dimensional construct comprising the individual's subjective perception of physical, emotional and social well-being, including both a cognitive component (e.g. satisfaction) and an emotional component (e.g. happiness) [7]. In addition to overall or global quality of life there are many specific sub-domains (e.g. health, job, family, friends, community, etc.). Some research on the impact of health on quality of life has examined the impact of domain-specific satisfaction on global life satisfaction. In this research, health-related quality of life has a modest effect on overall life satisfaction in the general population compared to satisfaction in other domains [8,9], but its effect is larger when focusing on those who experience major decrements in health status.

There has been substantial research on the effect of objective health status on overall life satisfaction or on a global measure of health-related quality of life. Yet, while the objective dimension of health status (as assessed by physicians' reports of symptoms or the presence of complications, for instance) is important, the patient's subjective perceptions of health translate the objective facts of his or her health status into an actual quality of life experience. This view is generally endorsed by researchers in this field [10–12], who point out that since expectations regarding health and the ability to cope with limitations and disability can greatly affect a person's perception of health and satisfaction with life, two people with the same objective health status may have a very different quality of life [5].

There is also general agreement that various domains of functioning and well-being can each contribute independently to global quality of life, thus making multidimensional measurement of quality of life necessary [10]. Simply asking one question, such as 'please rate your overall health-related quality of life on a scale from 0 to 100', may provide a useful global assessment, but it does not identify the underlying dimensions which contribute to the overall or health-specific quality of life [5]. Thus, almost all quality of life research involving people with diabetes employs multidimensional assessment of quality of life and typically assesses several dimensions, including physical, psychological, and social functioning and well-being.

Two broad approaches to health-related quality of life measurement have emerged – generic and disease-specific. The generic approach involves the use of measures applicable across health and illness groups. The most widely used generic measure of quality of life in studies of people with diabetes is the Medical Outcomes Study (MOS) Short-Form General Health Survey [13], in its several forms (SF-36, SF-20, SF-12). The MOS instrument includes physical, social and role functioning scales to capture behavioral dysfunction caused by health problems. Measures of mental health, perceptions of

overall health, and pain intensity reflect more subjective components of health and general well-being. The authors of this measure claim that these six health concepts are comprehensive in terms of those aspects of health considered most important to patients [14]. These instruments have been translated into many languages, and used in these forms in studies which include people with diabetes. The Rand Quality of Well-Being Self-Administered (QWB-SA) survey [15] is similar to the SF-36 in its aim to comprehensively assess health-related well-being or quality of life. It contains scales designed to measure acute and chronic emotional and physical symptoms, mobility, and physical activity. Other instruments used at least occasionally to assess general health status in people with diabetes include the Sickness Impact Profile [16] and the Nottingham Health Profile [17].

Generic measures like the SF-36 are most useful for comparing quality of life in people with different diseases and the quality of life in people who have no diseases with the quality of life in people who have a disease. Some generic measures, such as the Quality of Well-Being Scale [15], generate a single utility index of overall quality of life. This index usually ranges from 0 to 100 and these values can be used to adjust for years of life by degree of health experience to yield a measure of 'quality-adjusted life years'. Such a measure can be used to assess cost-effectiveness and cost benefits across various interventions and illnesses.

Many generic measures of emotional status have been employed in studies which include people with diabetes. These include the Well-Being Questionnaire [18], the Profile of Mood States [19], the Symptom Checklist (SCL-90R) [20], the Mini-Mental Status Exam [21], the Kellner Symptom Questionnaire [22], and the Affect Balance Scale [23]. Depression in people with diabetes has been studied using the following scales: the Beck Depression Inventory [24], the Zung Self-Rating Depression Scale [25,26], and the Center for Epidemiological Studies Depression Scale [27]. Anxiety in people with diabetes has been studied using the following scales: the Beck Anxiety Inventory [28], and the Zung Self-Rating Anxiety Scale [29]. Both depression and anxiety in people with diabetes have been studied using the Hospital Anxiety and Depression Scale [30].

Illness-specific quality of life measures can focus on the specific problems posed by an individual illness. For example, even a well-designed generic quality of life scale will not address certain aspects of life with diabetes such as hypoglycemia, insulin injections, self-monitoring of blood glucose (SMBG), and dietary restrictions, which may be critical to an individual's health-related quality of life. Generic measures may not be specific enough to detect effects in some areas of functioning among some people with diabetes. For example, generic measures of mental health may not identify fear of complications as an important contributing factor. More and more, researchers have added disease-specific assessments to generic ones, to increase the ability of their measures to identify the factors most relevant to the health-related quality of

life of people with a specific disease. Some [12] have even advocated a 3-level approach for clinical trials, incorporating generic measures, disease-specific measures and, finally, situation-specific questions that apply to the specific condition (neuropathy, for example) or intervention being investigated.

The most widely used diabetes-specific quality of life measure is the Diabetes Quality of Life (DQOL) measure [31], developed for use in the Diabetes Control and Complications Trial (DCCT). The DQOL was designed to measure diabetes-specific quality of life. It contains scales to assess five separate areas: satisfaction with treatment; impact of treatment; worry about the future effects of diabetes; worry about social and vocational issues; and overall well-being. The last scale was derived from national surveys of quality of well-being and can be used to compare people with diabetes and a wide variety of other populations. The Satisfaction and Impact scales seem to be broad gauges of diabetes-related quality of life, whereas the Worry scales address concerns more specific to patient perceptions of diabetes-related emotional distress.

Since the DQOL was introduced, a number of other comprehensive diabetes-specific quality of life measures have been developed. The Diabetes-39 instrument [32] includes scales to measure energy and mobility, impact of treatment, worries about the future effects of diabetes, worries about diabetes-related social issues, and sexual functioning. The authors of the scale developed it for use with people who have either Type 1 or Type 2 diabetes – whether managed with insulin, oral agents or diet alone. The Diabetes-39 has been translated from its original US English version into a wide variety of other languages, including UK English, Danish, Dutch, Finnish, French, German, Italian, Norwegian and Swedish.

The Problem Areas in Diabetes (PAID) survey [33] is a relatively new measure of psychosocial adjustment specific to diabetes. The PAID contains items measuring burden of illness, satisfaction with treatment, impact of treatment, and worries about the future effects of diabetes. The authors designed the PAID, which may be used with patients who have either Type 1 or Type 2 diabetes, to tap the breadth of *emotional* responses to diabetes.

Lewis and colleagues [34] developed an instrument, the Diabetes Treatment Satisfaction Questionnaire (DTSQ), designed to measure only diabetes treatment satisfaction.

Does quality of life differ among identifiable subgroups?

The literature concerning associations between quality of life and disease-specific, demographic and psychosocial variables in people with diabetes is larger than that for differences in quality of life between people with diabetes and the general population, or between people with diabetes and those with other chronic conditions. These

studies employ diabetes-specific as well as generic measures of health-related quality of life.

Quality of life and duration of diabetes

Here findings are mixed. Several studies found that increased duration of diabetes was associated with decreased quality of life, as assessed either by the Nottingham Health Profile in a Finnish population of people with each type of diabetes [35] or by the SF-36 and SF-20 in populations of people with both types of diabetes [36,37]. On the other hand, some have found no significant association between quality of life and disease duration. The present authors [38] reported no significant association between disease duration and depression in a population which included those with Type 1 and Type 2 diabetes; a Finnish study of patients with Type 1 diabetes found no significant relationship between disease duration and scores on the Finnish version of the SF-20 [35]; a Swedish study of people with Type 2 diabetes found no significant association between disease duration and scores on Bradley's Well-Being or Treatment Satisfaction questionnaires [39]; and a study of people with Type 1 diabetes found no significant relationship between disease duration and DQOL scores [40]. One study of Type 1 patients in the United Kingdom found that DQOL treatment satisfaction scores were actually higher for those who had diabetes longer [41].

Quality of life and the presence of diabetes-related complications

The research addressing this question is consistent in finding that the presence of complications, particularly the presence of two or more complications, is associated with worsened quality of life. In fact, this finding is so strong that it suggests that inconsistent findings with regard to the association between other variables and quality of life may be explained by the frequent omission of this factor as a possible confounding variable.

Studies conducted by the current authors found that the presence of two or more diabetes-related complications was associated with a significant increase in the likelihood that patients with either type of diabetes had clinically meaningful symptoms of depression or anxiety [38]. Other studies we conducted [42] indicated that the presence of complications was associated with increased DQOL treatment burden scores. Jacobson and his colleagues reported that lower scores on all SF-36 scales were associated with greater severity of complications for patients with either type of diabetes, and with number of complications among those who had Type 1 diabetes [43]. Among those with Type 2 diabetes, who had fewer complications, the number of complications was a weak predictor of SF-36 scale scores. A similar pattern of findings was reported for the association between number and severity of complications and scores on DQOL scales, with treatment satisfaction and disease impact scales consistently sensitive to severity of compli-

cations and less consistently responding to number of complications. Trief and colleagues [44] reported that number of complications was a strong predictor of DQOL diabetes impact and treatment satisfaction scores in a population of insulin-requiring patients. In addition, number of complications was associated with lower scores on the Nottingham Health Profile [45], and increased number of complications was associated with lower scores on all scales of the SF-36 [46] and the SF-20 [30].

Polonsky and colleagues found that PAID scores measuring psychosocial adjustment specific to diabetes, were positively associated with short-term and long-term diabetes complications [33]. Others have found that the presence of neuropathy, cardiovascular disease or end-stage renal disease was associated with decreased scores on all scales of the SF-36 [37,47]; the presence of end-stage renal disease (ESRD) was associated with markedly increased functional impairment as measured by the Sickness Impact Profile [48]; and the presence of nephropathy was associated with greater health worries and reduced perceived health in patients with Type 1 diabetes [40]. Another study examined quality of life in insulin-treated patients with gastroparesis (a complication of diabetes), and reported that quality of life in this population was significantly lower than the norms for those with Type 2 diabetes on all eight SF-36 subscales [49].

Several researchers have found increased depression and negative life experiences during the two years after diagnosis with proliferative diabetic retinopathy (PDR) [50,51]. These psychosocial disruptions existed regardless of the severity of the visual impairment and were maintained even after lost vision was regained. It has been estimated that 50% of diabetic men with impotence problems have a significant emotional overlay attributable to depression or anxiety that contributes to erectile dysfunction [52]. Others have found a significant association between sexual problems and depression among diabetic men [53,54] and women [55]. These psychological factors may both exacerbate and be exacerbated by organic pathology in the development and maintenance of sexual dysfunction [56].

Quality of life and type of diabetes

Jacobson and colleagues [43] used the SF-36 and the DQOL to assess quality of life in people with Type 1 and Type 2 diabetes, and found that Type 2 patients not taking insulin reported higher quality of life than Type 2 patients taking insulin. Type 2 patients on insulin still experienced better health-related quality of life than Type 1 patients. While it makes sense to think that the effects of Type 1 and Type 2 diabetes on quality of life may differ, it is unclear what the sources of these differences might be, other than factors associated with diabetes type, such as treatment regimen and age. In fact, some researchers, including the current authors, have found few meaningful differences between those with each type of diabetes in

functional status or well-being. One study found that levels of depressive symptomatology were not significantly different as a function of diabetes type [38], and another found little disruption in functioning or well-being for those with either type of diabetes [57]. Respondents in the latter study who had Type 1 diabetes reported some interference with their work lives, and those with Type 2 diabetes reported some interference with their social lives. In another study [58], those with Type 1 diabetes were similar to those with Type 2 diabetes on most SF-36 subscales, though those with Type 1 diabetes reported better physical functioning, more role limitations due to physical health, fewer role limitations due to emotional problems, more energy, less anxiety, and less favorable health perceptions. Based on the limited available data, it is probably fair to say that while quality of life or some of its components may differ as a function of diabetes type, these differences are probably the result of other factors, such as treatment regimen or age, which are associated with diabetes type.

Quality of life and treatment regimen

Results of research on the association between treatment regimen and quality of life in people with diabetes are mixed, with some indication that increasing treatment intensity in patients with Type 2 diabetes from diet and exercise alone, to oral medications, to insulin, is associated with worsening quality of life. Jacobson and colleagues reported that patients taking oral medications had more DQOL-assessed diabetes-related worries than those controlling their diabetes with diet and exercise only, and that those taking insulin reported less DQOL-assessed satisfaction with treatment and more burden of illness than those taking oral blood-glucose-lowering medication or none at all [43]. Similarly, others have found lower scores on Bradley's Well-Being and Treatment Satisfaction questionnaires in Type 2 patients treated with insulin compared to those who were not [59]; higher scores on all Nottingham Health Profile scales for Type 2 patients treated with diet only compared with those taking oral blood-glucose-lowering agents [45]; lower scores on SF-20 physical and social well-being scales for Type 2 patients treated with insulin compared with those who were not [36]; and lower scores on the Profile of Mood States vigor scale and in general well-being in patients switching to insulin therapy [60].

In contrast, one study of people with Type 1 diabetes conducted in the UK found no association of any DQOL scale with multiple versus twice-daily insulin injection regimes [41]. Another study found no significant association between type of therapy (diet, oral agents or insulin) and quality of life in people with Type 2 diabetes, as assessed by the Profile of Mood States and the Social Difficulties Questionnaire [57]. Other studies found no significant differences between patients with Type 2 diabetes who were treated with insulin and those who were not on Bradley's Well-Being and Treatment

Satisfaction questionnaires [61] or depression and anxiety [38].

Two studies examined the association between blood-glucose monitoring and quality of life. One found no association between SMBG frequency and scores on Bradley's Well-Being and Treatment Satisfaction questionnaires [39], and the other found no significant difference between elderly patients monitoring blood glucose and those testing urine glucose on a diabetes-specific quality of life questionnaire [61].

Quality of life and glycemic control

The past few years have brought a burgeoning of research on the relationship between glycemic control and quality of life in people with diabetes, and a number of these studies suggests that a relationship does exist, especially when quality of life is assessed by diabetes-specific measures rather than generic ones. Studies employing generic measures such as the SF-36, SF-20 or non-English-language versions of these instruments often reported null findings [46,47,50,62–65]. Only one study which used the SF-36 to assess quality of life found significant associations between HbA_{1c} and some SF-36 scales in some sub-populations. In the Wisconsin Epidemiologic Study of Diabetic Retinopathy (WESDR), Klein and colleagues [37] found that SF-36 general health and overall self-rated health scores were associated with HbA_{1c} levels for younger onset subjects only (i.e. diagnosed before 30 years and taking insulin).

Wikblad and colleagues [66] reported that scores on the Swedish Quality of Life Scale (SWEDQUAL) were lowest for those with the highest HbA_{1c} levels (>8.1%), highest for those with HbA_{1c} levels 7.1–8.0%, and intermediate for those with the lowest HbA_{1c} levels (≤7.0%). On the basis of these data, the authors suggest that there may be a curvilinear relationship between HbA_{1c} level and health-related quality of life, perhaps as a result of decrements in quality of life associated with more complex treatment regimens or increased incidence of hypoglycemia. Weinberger and colleagues [63] explicitly rejected this suggestion, stating that their own data, based on SF-36 responses, shows no linear or curvilinear relationship between HbA_{1c} and quality of life.

Others have used disease-specific measures to assess health-related quality of life in people with diabetes. In a study of 150 insulin-requiring adults [44], HbA_{1c} levels were significantly related to all DQOL scale scores, but to no SF-36 scale scores. Saudek and colleagues [67] found a significant association between HbA_{1c} levels and DQOL overall scores and disease impact scores, but not between HbA_{1c} levels and DQOL treatment satisfaction or diabetes worry scale scores, in a population of men with Type 2 diabetes taking insulin. Others have reported that those with lower HbA_{1c} levels had higher diabetes treatment satisfaction scores [68], lower scores on the PAID survey [33], and higher scores on Bradley's diabetes-specific well-being scale [69]. A few studies have found no significant relationship between HbA_{1c} levels and dia-

betes-specific measures of quality of life [39,59], but the HbA_{1c} levels of the participants in these studies were quite low, averaging about 7.0%, so the restricted range of glycemia may have contributed to the null finding.

Some studies have also reported lower levels of emotional disturbance in those who have lower HbA_{1c} levels. Van der Does and colleagues [70] found that worse mood (as indicated by displeasure, depression, tension and fatigue scores) was associated with higher levels of HbA_{1c} in a Dutch population of patients with Type 2 diabetes. Mazze and colleagues [71] found lower levels of depression and anxiety in subjects with Type 1 diabetes who were in good control when compared with those in average or poor control. In contrast, a study by the present authors [38] found that those with lower HbA_{1c} levels were not significantly less likely to report levels of depression and anxiety symptoms indicative of a clinical disorder, though there was a tendency in this direction.

Some studies have found significant associations between quality of life and measures of glycemia other than HbA_{1c}. Lower fructosamine levels were associated with higher DQOL treatment satisfaction scores [41], lower fasting plasma glucose levels were associated with lower levels of fatigue as measured by the Profile of Mood States [57], and fewer self-reported symptoms of hyperglycemia were associated with reduced diabetes burden and increased treatment satisfaction [32].

Overall, the preponderance of studies suggest that better glycemic control is associated with better quality of life. This association is stronger for measures of diabetes-specific quality of life and generic measures of emotional distress than for generic measures of quality of life. Although some studies fail to reveal this association, they do not counterbalance the instances of significant relationships. It appears that the benefits of good glycemic control more than offset the increased burden it may involve, at least for the majority of patients.

Quality of life and gender

A number of researchers have reported that quality of life is better among diabetic men than among diabetic women. This is consistent with reported gender differences in health-related quality of life in the general population [72–77]. We found [78] that men were more satisfied with their diabetes treatment regimen, and missed less work and fewer leisure activities as a result of their diabetes, than women did. We also found [42] that treatment satisfaction was higher and diabetes burden lower in men than in women, and [38] that men were significantly less likely to report symptoms of depression or anxiety consistent with the presence of a clinical disorder than women. Others have found that men with diabetes report less disease impact [41,79], more treatment satisfaction [39,59,79], and higher scores on all SF-20 scales [36] than women. These findings, suggesting that diabetic men have an advantage over diabetic women in health-related quality of life, reinforce

the need to control for gender in future investigations of quality of life in diabetes.

Quality of life and demographic variables

While we [38] have found no meaningful pattern of association between age and quality of life, others who assess aspects of functioning more likely to be affected by age suggest there is an association between age and specific aspects of well-being. As part of a large, national diabetes self-management survey, Glasgow and colleagues [36] found that younger persons had significantly higher scores than older persons on SF-20 scales measuring physical functioning and social functioning, and lower scores than older persons on the SF-20 scale measuring mental health. Similarly, Klein and colleagues, using the SF-36 as part of the WESDR [37], found that older subjects reported lower levels of physical functioning and physical role functioning than younger subjects. It appears that advancing age does affect some aspects of health-related quality of life, especially those associated with physical functioning, in people with diabetes, so an accurate assessment of the association between diabetes and quality of life must control for age.

Significant associations have also been demonstrated between socioeconomic status (measured by income or educational level) and quality of life in the general population [80–82]. Similarly, we [38] found that study subjects who graduated from college were significantly less likely than those with less education to report symptoms of depression or anxiety consistent with the presence of a clinical disorder. Finally, Glasgow and colleagues [36] reported that survey respondents who reported more education and higher income also scored higher on all sub-scales of the SF-20. Thus, it appears that socioeconomic status should be controlled for in definitive studies of the relationship between diabetes and health-related quality of life.

Few have studied the relationship between race or ethnicity and quality of life in people with diabetes. We found no association between race and measures of anxiety or depression when other demographic and disease factors were controlled [38], and Glasgow and colleagues found no differences between Caucasian and African-American respondents on any dimension of quality of life as measured by the SF-20. One study [82] found that European-Americans scored higher than Chinese immigrants to the United States on all DQOL scales. While race and ethnicity do not appear to be significantly associated with quality of life among people with diabetes, further research would help to clarify the nature of this relationship.

Marital status appears to be related to quality of life in the general population [80,83], and we [38] found that study subjects who were not married were significantly more likely than those who were married to report symptoms of depression consistent with the presence of a diagnosis of clinical depression. Jacobson and colleagues

[43] reported a pattern of relationships between marital status and quality of life (as measured by the SF-36 and DQOL), which indicated that separated or divorced individuals experienced worse quality of life than those who were single or married. A study of people with Type 2 diabetes conducted in Norway found that those living alone reported lower levels of physical functioning and psychosocial well-being than those who lived with others [84].

Quality of life and psychosocial predictors

Some have suggested that health-related quality of life in people with diabetes may be affected by psychosocial factors such as health beliefs, social support, coping strategies and personality traits. For example, Rose and colleagues [10] found that subjects who reported feeling more socially competent, who received more practical support for diabetes management, and who coped more actively, reported higher levels of functioning and well-being as well as higher levels of global life satisfaction. Using a sophisticated statistical modeling technique, the authors found that coping strategies and personality traits significantly covaried with all quality of life realms, and had a greater effect than the presence of complications on overall health-related quality of life.

Similarly, we [85] found that subjects with an internal-autonomous diabetes locus of control orientation, people who believed they could effectively manage their diabetes, were less depressed and anxious than those with other diabetes locus of control orientations. In the same study, we reported that those with a chance diabetes locus of control orientation, subjects who believed that diabetes control was a matter of chance or fate, were more depressed and anxious, and had lower levels of self-esteem than those with other diabetes locus of control orientations. Other researchers have reported that higher levels of perceived social support were associated with higher levels of social functioning in diabetic patients on intensive insulin treatment [86]; higher levels of social support were associated with higher overall quality of life in patients with Type 1 diabetes [87]; higher self-reported levels of self-efficacy and diabetes-related social support were associated with higher scores on the Finnish version of the SF-20 in a group of patients with Type 1 diabetes [35]; and better social relations and fewer family arguments were associated with better health-related quality of life as assessed by the Duke Health Profile and the General Health Perceptions Questionnaire [40].

Quality of life differences in people with and without diabetes

This question can be answered in two ways. First, people may be compared who are similar except that one group has diabetes while the other does not. Most often people with diabetes are compared to the general population,

sometimes to persons without any chronic disease, and sometimes to persons who have the same additional diagnoses other than diabetes. Secondly, persons with diabetes may be compared to persons without diabetes who have another specified diagnosis. The first approach assesses the additive effect of diabetes while the second approach assesses the effect of diabetes relative to other disorders.

Does diabetes itself affect quality of life?

In general, most studies report that quality of life among people with diabetes is worse than quality of life in the general population. Several studies measured quality of life using the SF-36, the SF-20, the SWEDQUAL (a Swedish scale based on the SF-36), or a Finnish version of the SF-20. Ware and colleagues published data based on responses to the 1990 National Health Survey of Functional Status [14,88–90], which included a sample of 541 people with Type 2 diabetes, with a mean age of 60.2 years, 55.6% of whom were women. These researchers found that those with diabetes reported lower quality of life than the general population on scales assessing physical functioning, role functioning and general health perception, but differences were not significant on scales measuring social functioning and mental health. A Swedish study which did not examine subgroups of diabetic subjects (as defined by disease characteristics) reported that quality of life was higher for controls than for those with diabetes on all 12 quality of life scales except social health [91]. A Finnish study which included patients with both types of diabetes but did not distinguish subjects by complication status found that those with Type 2 diabetes scored lower than controls on all quality of life scales, while those with Type 1 diabetes scored lower than controls on all scales except mental health [92].

A study of people with Type 1 diabetes reported that Rand General Well-Being Questionnaire scores for health-related functioning were lower in this group than in controls, but that scores for functional limitations and physical disabilities were not significantly different [93]. A French study, which did not specify the type of diabetes or complications status of its subjects, found that those with diabetes had lower quality of life than controls on most scales, but not on measures of cognitive functioning [94]. Scandinavian studies which included people with both types of diabetes found that these subjects reported lower well-being [69] and more illness-related absences from work, less satisfaction with their leisure time, and fewer social contacts [95] than controls.

Several studies have compared those with a common diagnosis, some of whom also have diabetes. Among a group of patients enrolled in a cardiac rehabilitation program following a major cardiac event, those with diabetes were more depressed and reported poorer general health perceptions and well-being than those who did not have diabetes [96]. Among patients with a

first episode of severe limb ischemia, those with diabetes had poorer quality of life in all assessed aspects than those who did not have diabetes [97].

Studies of quality of life employing instruments other than the SF-36 or its derivatives also suggest that life may be more difficult for some segments of the diabetic population than for those who have no chronic conditions. A recent German study which included people with both types of diabetes [10] found that quality of life did not differ between those with and without diabetes, except when those with diabetes had complications.

A number of studies suggest that people with diabetes have higher levels of psychological disturbance, especially depression and depressive symptomatology, than norms derived from the general population. Researchers who do not obtain prevalence estimates for sub-samples of their diabetic subjects generally find that depression is more common among those with diabetes than among the general population [93,94,98–100]. A study [38] by the present authors found that rates of disturbance for depression (41%) and anxiety (49%) were higher than those typical in the general population (<10%). However, we also found that probability of disturbance was strongly associated with the number of diabetes-related complications and with certain demographic variables, especially gender and level of education. Among those without any of these additional risk factors, the risk of depression was quite low. These findings reinforce the point that certain disease and demographic characteristics may powerfully affect quality of life in people with diabetes, while diabetes per se may not.

Quality of life in people with diabetes and in people who have other chronic conditions

Quality of life in people who have diabetes has been compared with that in people with a variety of other chronic conditions and diseases, including hypertension [13,58,62], congestive heart failure [13,58], myocardial infarction [13,58], heart disease [62,96], angina [69], stroke [69], epilepsy [62], chronic hepatitis C [10], inflammatory bowel disease [10], and depression [13,96]. A few of these studies distinguish between diabetic subjects who have Type 1 and Type 2 diabetes, and between those with complications and those without, but most studies do not make these distinctions. The latter is problematic because the presence of complications is associated with poorer quality of life.

Stewart and colleagues [13] evaluated the functioning and well-being of 9385 adults at the time of office visits to 362 physicians in three US cities, using the MOS SF-36, and found that patients with hypertension had scores on all SF-36 scales similar to those for subjects with no chronic conditions. Patients with diabetes had lower scores than controls on scales assessing physical, role functioning, social functioning and health perceptions, but not on scales measuring mental health or bodily pain. Heart disease (congestive heart failure (CHF), angina,

and recent myocardial infarct (MI)) negatively affected physical, role and social functioning, and mental health (but not health perceptions or bodily pain, except for those with angina) to a greater degree than diabetes, and gastrointestinal problems negatively affected perceptions of mental health and bodily pain (but not other elements of functioning or well-being) to a greater degree than diabetes. Other conditions, including arthritis, chronic lung problems and back problems, were associated with SF-36 sub-scale scores similar to those for diabetes, except that physical, role and social functioning and mental health scores were lower for those with chronic lung problems, health perception scores were higher for those with arthritis, and bodily pain scores were higher for those with arthritis and back problems. The authors also note that patients with multiple conditions show greater decrements in functioning and well-being than those with only one condition.

In another study [58] some of the same authors compared quality of life in those with Type 1 diabetes, Type 2 diabetes, hypertension, congestive heart failure, recent myocardial infarction and depression, and reported findings generally consistent with their earlier research. People with Type 1 diabetes had SF-36 functioning and well-being scores similar to those for people with hypertension on all scales, except those assessing role limitations due to physical health and current perceptions of health, on both of which those with Type 1 diabetes reported poorer quality of life. Study respondents with CHF, compared with those with hypertension or either type of diabetes, reported lower quality of life on a number of measures, including physical functioning, role limitations due to physical health, energy (similar to Type 2 diabetes), sleep and current health perceptions (similar to Type 1 diabetes). Respondents who were depressed reported lower levels of functioning and well-being compared to those with diabetes on almost all measures except energy (similar to Type 2 diabetes) and current health perceptions (similar to Type 1 diabetes).

In contrast to the findings of these studies, Testa and Simonson [101] compared subjective assessments of mental, emotional and perceived health in individuals with no chronic condition, those with diabetes, and those with hypertension, and found that subjects with diabetes scored consistently higher on all scales than their hypertensive counterparts and slightly lower than controls. The authors suggest that the quality of life advantage of the diabetic patients in this study may result from the fact that all but one of the hypertensive patients were taking a variety of medications, including diuretics and beta blockers, which might significantly affect quality of life. In addition, the diabetic study subjects were all participants in the Joslin Diabetes Center education program, which might indicate that these subjects were unusually well motivated or well prepared to manage the stresses of life with diabetes.

In another study using the SF-36, functioning and quality of life for those with diabetes (type unspecified)

was poorer than those with hypertension or epilepsy (seizure-free), better than those with heart disease or epilepsy with seizures, and much better than those who were depressed [62]. The SF-36 was also used in a study comparing health-related quality of life in patients with diabetes, epilepsy and multiple sclerosis [102]. Patients with diabetes or epilepsy scored similarly and higher than those with multiple sclerosis on scales assessing physical functioning, role limitations due to physical health, social functioning and energy. Patients with diabetes scored lower than those with epilepsy or multiple sclerosis on the scales measuring emotional well-being and role limitations due to emotional problems, and those with diabetes or multiple sclerosis reported poorer health perceptions than those with epilepsy.

In a study [69] of the entire adult population of a Norwegian county (total number of participants = 74 977), well-being was assessed by means of an 8-item questionnaire, and scores for those with diabetes, myocardial infarction, angina and stroke were compared with those who had no chronic condition. Quality of life for those with diabetes was poorer than for those with no chronic conditions, but better than for those with angina or stroke.

In a German study [10], patients with diabetes under intensive insulin therapy were studied, as were patients with inflammatory bowel disease, chronic hepatitis C, and control subjects. Those with diabetes who had no complications were similar to controls on all measures of quality of life, and reported better functioning and well-being than those with any of the other chronic diseases considered. Those with diabetes who did have complications reported quality of life worse than that of controls and about the same as that of subjects with other chronic conditions.

Most studies report that quality of life is worse for people with diabetes than in the general population, especially with regard to physical as opposed to social or mental aspects of well-being. Most studies report that quality of life is better for people with diabetes than for people with most other chronic conditions, with the possible exception of hypertension. At the same time, most studies do not include and generate prevalence estimates for sub-samples of diabetic subjects who vary by disease characteristics (notably the presence and number of complications), or demographic characteristics (especially gender and education or socioeconomic status), which appear to be strongly associated with quality of life. Studies which look more closely at specific domains of functioning and well-being in people with each type of diabetes suggest that Type 1 diabetes may be associated with decrements in role limitations due to physical health and current health perceptions, while Type 2 diabetes (perhaps partly as a function of the more advanced age of this group) may be associated with decrements in physical functioning, role limitations due to emotional problems, and energy level. Because most studies do not generate estimates for sub-samples of diabetic subjects who vary by disease or demographic

characteristics which are strongly associated with quality of life, it is not possible to conclude that quality of life differences are due to diabetes per se rather than some other characteristic associated with diabetes. Nor is it possible to conclude which subgroups of diabetes patients have better or worse quality of life than non-diabetic comparison groups.

Can quality of life in people with diabetes be improved?

Several studies describe medical interventions designed to improve health status in people with diabetes, and report assessments of impact on quality of life. One study [60] reported that patients who had a decrease in HbA_{1c} of 1% or more during the one-year course of a clinical trial tended to have comparatively favorable mood (especially displeasure scores) and general well-being scores at follow-up. In a randomized, controlled, double-blind trial in which participants took either glipizide GITS or placebo, Testa and colleagues [11] assessed a variety of outcomes, including quality of life, and found that increases in HbA_{1c} levels of $\geq 1\%$ were associated with substantial decrements in quality of life, while decreases of the same magnitude showed smaller, but clinically relevant, improvements in quality of life. In their efforts to explain the ability of their study to detect a strong relationship between glycemic control and quality of life when others found none, the authors point to several possible explanations. They note that participants in their study experienced very little hypoglycemia, thus creating no offset to the quality of life improvements seen with reduced hyperglycemia. The authors also note that their placebo control group had substantial hyperglycemia, thus increasing the likelihood that any glycemia-related effects on quality of life could be detected. Another study, of patients with diabetes and painful neuropathy, found that patients treated with gabapentin reported significantly improved scores on all SF-36 and Profile of Mood States (POMS) sub-scales, as compared with placebo-treated subjects [103].

A study of insulin-treated patients with gastroparesis [49] found that those who experienced symptomatic improvement of their gastroparesis also demonstrated significantly greater improvement than domperidone non-responders on all SF-36 scale scores but one (physical functioning). Two uncontrolled studies found that patients switching from syringes to pens as a means of insulin delivery reported improved health-related quality of life [104] and heightened satisfaction with treatment [105]. Another study [96] found a 67% reduction in symptoms of depression in diabetic patients who completed a course of cardiac rehabilitation.

The DCCT assessed quality of life in both intensively and conventionally treated subjects with Type 1 diabetes over the course of their participation in the trial (average = 6.5 years), and found no significant differences on any measure of quality of life (DQOL, SCL-90R or SF-36) at any assessment [31]. These findings may be

seen as evidence that intensified treatment did not improve quality of life, or as evidence that a demanding intensive regimen which led to increased weight gain and hypoglycemia did not lead to decrements in quality of life. One must keep in mind that participants in the DCCT were selected for their motivation, commitment and ability to successfully manage the rigors of a demanding clinical trial of long duration. It also should be recognized that the intensive treatment group received extensive psychosocial support in order to enable them to maintain the demanding treatment regimen and remain in the trial.

In a related study of intensified insulin therapy in patients with Type 1 diabetes conducted in Switzerland [106], patients reported improvements from baseline to 12-month follow-up in global life satisfaction, time free from diabetes-specific strain, and anxiety and depression. Another study of intensified insulin therapy conducted in Germany [107] found that patients who increased the number of insulin injections they took each day or switched from injections to CSII reported increased DQOL treatment satisfaction and reduced diabetes burden.

A number of researchers have examined the effects of counseling and educational interventions for people with diabetes, and some studies have incorporated measures of quality of life or some of its elements. We [108] reported on a 5-day outpatient education program which incorporated a coping skills training intervention designed to improve quality of life, and found that participants reported significantly improved levels from pre-program to 6-month follow-up in self-esteem, depression and anxiety; at 12-month follow-up improvements in self-esteem and anxiety were maintained, but improvements from baseline in depression, though still substantial, were no longer significant. Anderson and colleagues [109] conducted a randomized wait-list control trial in which the treatment group received a 6-week (one session per week) patient empowerment education program. Treatment group participants showed gains over the control group on measures of diabetes impact and negative attitude toward diabetes. Other group interventions designed to facilitate psychosocial coping with diabetes have yielded improvements in coping skills and/or emotional well-being [110–113].

Thus, it appears that health-related quality of life in people with diabetes can be improved by certain medical interventions and by educational and counseling interventions designed to enhance coping skills. However, it generally is difficult to know what aspect of the intervention is producing the change in quality of life because all relevant factors are not measured and incorporated into the analysis.

Conclusions

Interest in health-related quality of life has burgeoned in recent years, fueled in part by growing evidence that psychosocial factors can powerfully influence self-man-

agement behaviors, and that some psychosocial variables may predict medical outcomes such as hospitalization or mortality to a greater degree than physiologic or metabolic variables. Recent changes in fields of health outcomes research have also contributed to interest in quality of life issues, bringing more attention to the assessment of short-term patient outcomes, including functional health status, satisfaction with health care, and overall quality of life.

Our review reveals several patterns in the existing research:

1. Compared to persons without diabetes, most studies reported worse quality of life for people with diabetes, especially regarding physical functioning and well-being. Those with diabetes report better quality of life than people who have a variety of other chronic conditions (such as cardiac problems, epilepsy with seizures, arthritis, multiple sclerosis, stroke and lung problems) in some or most domains of functioning and well-being. The domain-specificity of these findings must be kept in mind.
2. Type of diabetes does seem to be associated with quality of life when one looks at different domains of functioning and well-being, but these differences are likely the result of factors associated with type of diabetes, such as treatment regimen or age, and not a direct function of type of diabetes per se. A similar conclusion can be made with reference to duration of diabetes.
3. Findings regarding the relationship between treatment regimen and quality of life appear to differ by type of diabetes. For those with Type 2 diabetes, treatment intensification from diet only to oral agents to insulin seems to be associated with decrements in quality of life. For those with Type 1 diabetes, the picture is more complex. Some studies, such as those conducted as part of the DCCT, suggest that treatment intensification has no effect on quality of life; others suggest that such intensification improves quality of life by reducing acute and long-term complications of hyperglycemia; and still others suggest there may be a curvilinear association between treatment intensity and quality of life, since intensification may bring an onerous regimen or significant increases in hypoglycemia.
4. Quality of life is better in those with better long-term glycemic control, especially if quality of life is assessed using disease-specific measures which capture patient perceptions of symptoms, and as long as lower levels of glycemia are not associated with significant increases in treatment burden and hypoglycemia.
5. Some demographic variables associated with quality of life in people with diabetes parallel those in the general population. Specifically, men seem to report better quality of life than women; increasing age seems to be associated with decrements in some domains of functioning and well-being; and those with more education or income generally report better quality of life than those who have less of either of these attributes.

6. Some psychosocial factors, including health-related beliefs, social support, coping style and personality type, may powerfully affect quality of life, either directly or via their capacity to buffer the negative effects of diabetes or its management. In fact, it appears that these factors may predict quality of life more powerfully than important disease-related variables such as complications.

7. Quality of life in people with diabetes can be improved by certain interventions, including the introduction of blood-glucose lowering agents, changes in insulin delivery systems, and educational and counseling programs designed to facilitate the development of diabetes-specific coping skills.

We have found strong evidence for the power of disease-specific, demographic and psychosocial factors to influence functional status and well-being in people with diabetes. These considerations have substantial scientific and clinical implications. From a research perspective, the strong covariance between a variety of physiologic, behavioral, attitudinal, and demographic parameters and quality of life in people who have diabetes makes monocausal hypotheses or statements about general quality of life of limited value [10]. Future research should control for or explicitly assess the effects of multiple factors and not simply ignore them or treat them as an undesirable source of variance.

Researchers are striving to address a number of methodological issues in their efforts to study quality of life in people with diabetes. Several researchers have raised issues concerning the form of individual quality of life items used to generate global scores. Testa *et al.* [11] suggest that symptom distress is especially important in diabetes research. They hypothesize that it is the *bothersomeness* of a particular complaint or symptom, and not just its presence or even its frequency that determines quality of life [5,114]. Testa and her associates [5,11] have also written extensively on the importance of sensitivity to change. This work has emphasized that the clinical meaningfulness of quality of life differences cannot be determined by reference to the absolute magnitude of a change relative to the maximum possible range of the measure. For example, although the physical functioning scale of the SF-36 has an absolute range of 0–100, a decrease in physical functioning of greater than 5 units may have serious adverse effects on quality of life.

One major methodological issue has not been addressed by researchers in the field. It is unclear whether several of the factors examined affect quality of life through objective physical health or through subjective perceptions. For example, does enhanced glycemic control affect quality of life by reducing symptoms of hyperglycemia or by reducing fear of complications and enhancing perceived life expectancy? Much the same argument could be made for all disease-specific factors. Moreover, even the controlled clinical trials have not explicitly considered this question. Doing so would require blinding a subgroup of intervention

subjects to their glycemic control to determine whether effects of glycemia differ from subjects who know their glycemic status. Both subjective and objective influences may be at work, but we do not know their relative contributions to quality of life.

The current review clarifies the value of disease-specific and even complication-specific [115] or intervention-specific measures of quality of life, since more generic measures may lack the ability to identify the potentially important benefits of new therapeutic interventions, or the pathways through which these benefits are achieved. Utilizing scales to assess quality of life which are able to capture meaningful differences between groups and meaningful changes as the result of an intervention has profound practical implications. Quality of life research in diabetes will realize its full potential when it is able to answer clinically relevant questions, such as:

- What is the effect of a treatment program on day-to-day quality of life and compliance?
- Are the benefits of a treatment program offset (or enhanced) by treatment-related changes in quality of life?
- How can we design, implement and evaluate interventions which influence factors that may affect quality of life, including diabetes-specific beliefs and attitudes, social support, and disease-specific coping skills?

Coping skills may be a critical factor in this process [10], and one worth special attention because some have shown that these skills can be developed or enhanced through training [108–113]. Active, constructive coping may be seen as one element in a positive feedback loop, in which active coping leads to improved functional status and well-being, which leads to improved diabetes self-management, better physiologic outcomes, and finally to reinforcing effects on quality of life and an active approach to coping. This suggests that people with diabetes who are not active copers may require special attention [85], to help them from becoming trapped in a negative feedback loop which operates in the opposite direction from those who cope actively. In these patients, a sense of failure, depression, denial or passivity leads to reduced quality of life, reduced self-management efforts, poorer medical outcomes, and deeper feelings of failure, guilt and depression. Improving the capacity of these patients to cope may reverse the valence of this feedback loop from negative to positive.

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